Image Classification on Arduino Board

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I. Introduction

In this project, we implement an image classification model on Arduino Nano 33 BLE Sense board with OV7675 camera. The model is trained on Edge Impulse platform [2], aiming at classifying the letter in an image, which could be any one of the three (A, B, C). The letters in different images have different natures such as format (printed or handwritten), font, size, color, brightness, etc. The model reaches a validation accuracy of 95.8%, and is available for transferring to Arduino IDE and Nano 33 BLE board while keeping the accuracy.

The following steps are taken to implement the model and transfer it to Arduino board. (1) We prepare the images and upload them to the Edge Impulse platform. (2) We train an image classification model using the template Edge Impulse provides, which involves image squashing to reduce the image data size. (3) With the model reaching a satisfying accuracy, we download the deployment library with model data provided by Edge Impulse, integrate it with the Arduino firmware in C++ [1], debug and modify the integrated firmware in Arduino IDE, and upload it onto the Arduino board ready for real-time image capturing and classification.

II. Dataset

The dataset contains images that have one of the three target letters, namely “A”, “B” or “C”, and the label corresponding to the letter. Those images are taken with the OV7675 camera on the Arduino board, from printed letters on books and posters as well as handwritten letters on papers and phones, with a goal of diversifying those letters in such aspects as font, shape, color, brightness, etc. The images are 176*144 in size with data representing RGB colors. Those images are then uploaded to Edge Impulse via uploading the Edge Impulse-provided firmware [1] to the Arduino board and then connecting the board to the platform. The instructions for connecting to Edge Impulse platform can be found in “Connecting Arduino Board and OV7675 Camera to Edge Impulse” section of “Project Notes” document. Example images and their labels are shown in Fig. 1.

Fig. 1: example images uploaded to Edge Impulse platform, with their labels.

120 images are uploaded in total as the dataset, with each of the three labels corresponding to 40 images. They are uploaded as the training set for the model. While we do not need testing set for the time being, there will be testing images generated through real-time image classification tests on Edge Impulse.

III. Model Training on Edge Impulse

With image data uploaded to Edge Impulse, we then build and train the image classification model with the template Edge Impulse provides in "Create Impulse" page. [2] There are several settings in the platform, including image size, resize method, output features, etc. A view of the Create Impulse page is shown in Fig. 2.

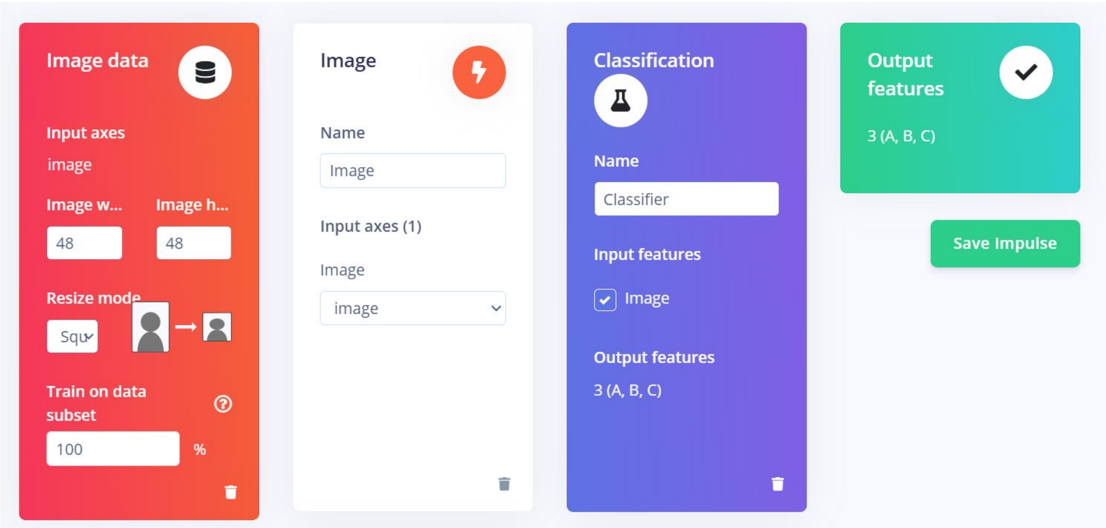

Fig. 2: view of Create Impulse page in Edge Impulse.

Then in the "Classifier" page, classifier-related settings, including number of training cycles, learning rate, training processor, neural network structure, etc. are provided. [2] The neural network used for the model contains two 2D convolution layers, one flattening layer, and a dropout layer with dropout rate of 0.25. We train the model for 30 epochs on CPU.

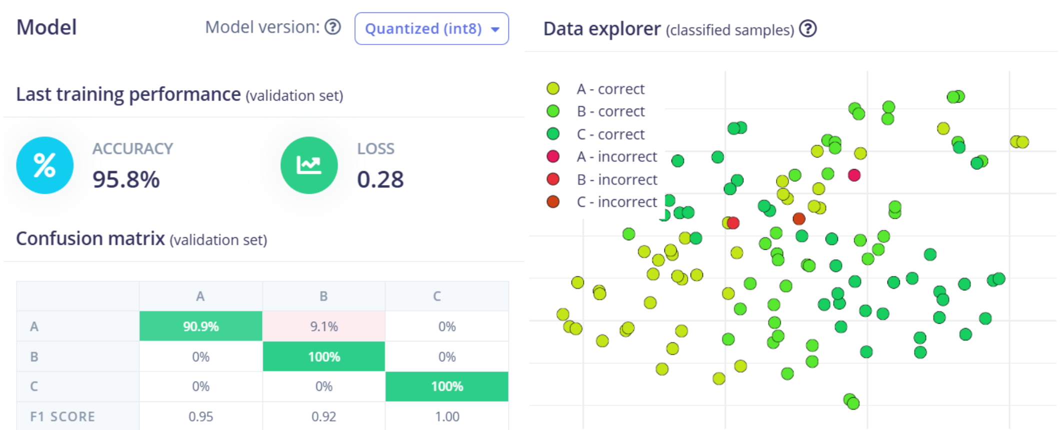

Fig. 3: training results of the model. The left shows the accuracy, loss, and classification vs. ground truth comparisons. The right shows the classified data as in the trained embedding space.

After the model is trained, the performance of the model is shown in Fig. 3. We can see that the model reaches an accuracy of 95.8% with our training data, which we are satisfied with. Hence, the next step is to export the model data and transfer them to Arduino IDE and Arduino board.

IV. Firmware Uploading to Arduino Board

For transferring the model to the Arduino board, we first download the deployment library with the trained model in “Deployment” page. [2] The library can be integrated with the Edge Impulse Arduino firmware in [1], through the instructions given in "Model Deployment on Arduino IDE" section in “Project Notes” document. Then, the integrated firmware is openable in Arduino IDE. After clearing a few debugging and file and library-related issues, which are detailed in “Problems and Solutions” section of “Project Notes” document, the firmware is able to be uploaded to the Arduino board. The complete, ready-for-uploading firmware can be found in [3].

We can then capture images by holding the Arduino board with OV7675 camera, and typing ‘c’ in Arduino IDE as instructed by the debug message. Note that we reduce the input image size for the model to 48*48 from 176*144, in order to fit the memory space of the Arduino board. Hence, it is suggested that when taking images with the Arduino board and OV7675 camera, hold it a bit further and to the lower right of the object targeted for capturing, as the image will be cropped with upper left part remaining. Users can visualize the input image using the ipynb image viewer in [4] by copying the data output printed in Arduino IDE, with instructions given in “OV7675 Camera Test” section of “Project Notes” document.

References

[1] Edge Impulse Firmware Library for Arduino. https://github.com/edgeimpulse/firmware-arduino-nano-33-ble-sense

[2] Edge Impulse Platform for Arduino Image Classification Project. https://studio.edgeimpulse.com/studio/738905

[3] Xiaohe Tian. Image Classification Model Firmware for Arduino IDE. https://github.com/xiaohet/arduino_image_classification

[4] TinyMLx Image Viewer. https://colab.research.google.com/github/tinyMLx/colabs/blob/master/4-2-12-OV7675ImageViewer.ipynb#scrollTo=WSozbpyAFs8a